TIAGo Pro

THE NEXT GENERATION OF **MOBILE MANIPULATORS**

BARCELONA | **2025**

TIAGo Pro

The Next Generation of Mobile Manipulators

TIAGo Pro Combines:

- NAVIGATION
- MANIPULATION
- PERCEPTION

To Enable:

- Mobile Manipulation
- Embodied AI
- Human-Robot Interaction

For R&D in:

- Healthcare
- Industry
- Agrifood

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TIAGo Pro

Designed to adapt to your research needs

HEIGHT
120cm - 155cm height *(fully extended torso)*

SEA ARMS
2x 7 DoF with 96cm reach
Fully Torque Controlled

SAFETY & VERSATILITY
Safety brakes, collision-aware navigation, series elastic actuators

OPERATING SYSTEM
100% ROS 2 integrated and **Open Source**

HRI AND EMBODIED AI
Designed to enable Human-Robot Collaboration and advanced AI research in human-shared spaces

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Extended Capabilities

Spatial flexibility for diverse setups and environments

Series Elastic Actuator (SEA) arms

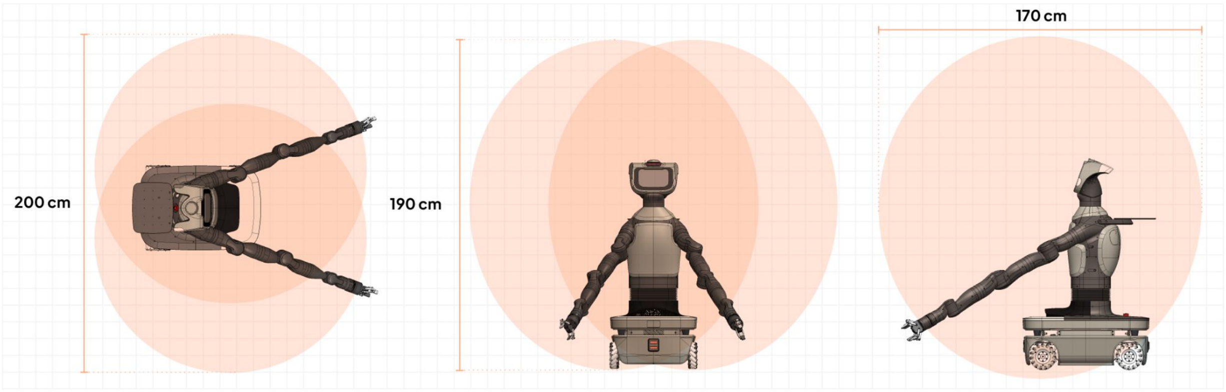
Lifting Prismatic Torso

Omnidirectional Navigation

- **Floor-level access** for extended task coverage
- **Effortless shelf reach** for improved handling
- **Seamless multitasking** across various applications

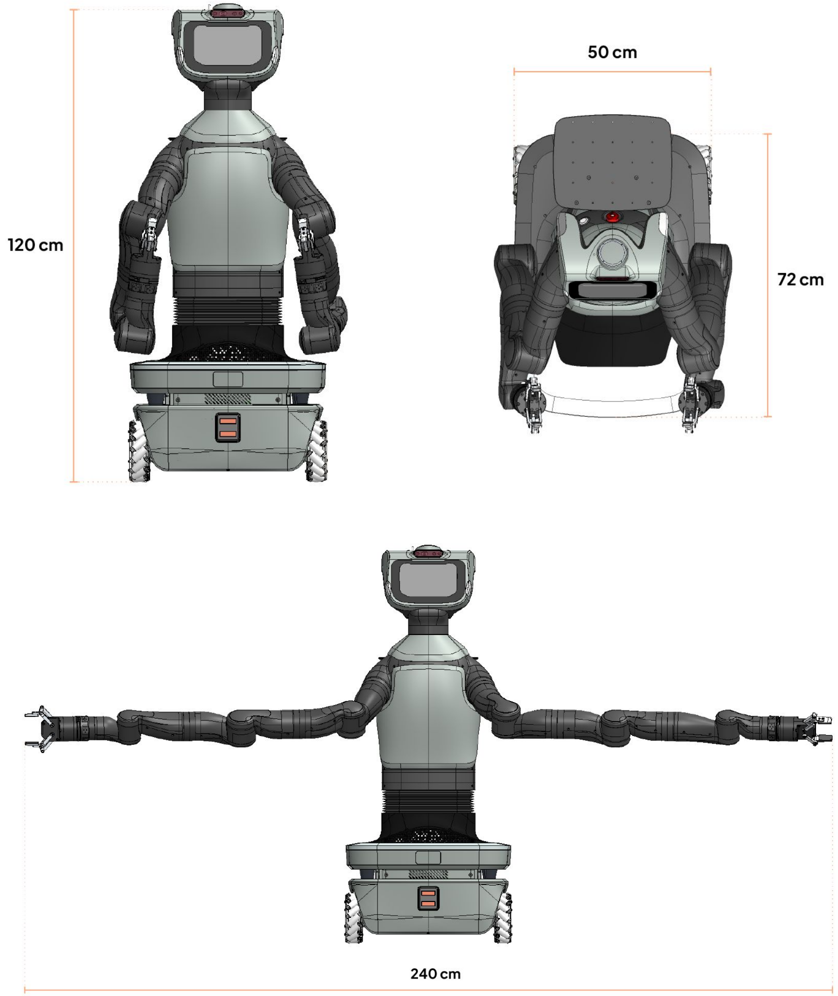
Enhanced **productivity, flexibility, and efficiency in mind** – both for research and real-world environments.

TIAGo Pro **Workspace:**



WITHOUT GRIPPER

TIAGo Pro **Measurements:**



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Advanced Design and Actuation

For controlled, responsive manipulation

Custom Series Elastic Actuator (SEA) **arm design enables safe interaction**

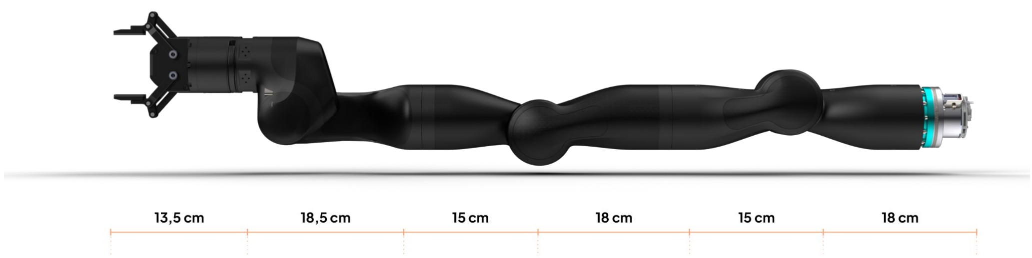
- **Absorbs energy upon impact,** reducing the risk of damage during collisions
- Integrated **safety brakes and force sensing in each joint** ensure controlled manipulation
- Safety mechanisms like the **emergency stop and dead man button** prioritize user protection during close interaction

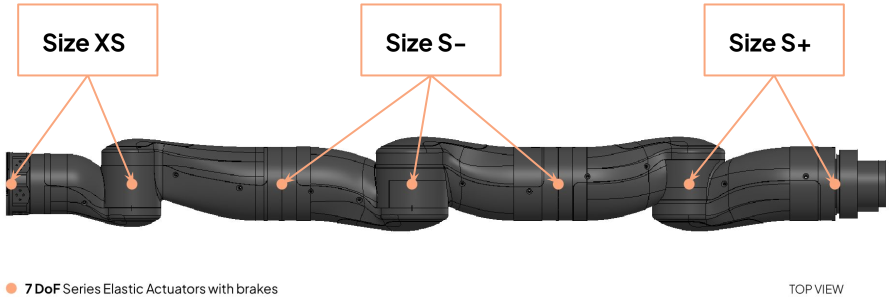
Robust, and versatile manipulation for developing, testing, and validating advanced research

Custom actuator design targets:

- **Precise motion** thanks to robust torque controllability
- **Flexible configuration** thanks to a modular design
- **Advanced, versatile control** thanks to precise, real-time communication

Without Force Sensor





Actuator size	Cont. Torque [Nm]	Peak Torque [Nm]	Peak Speed [rpm]
Size S+	45	70	35
Size S-	26	45	60
Size XS	20	30	60

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Safety and Control

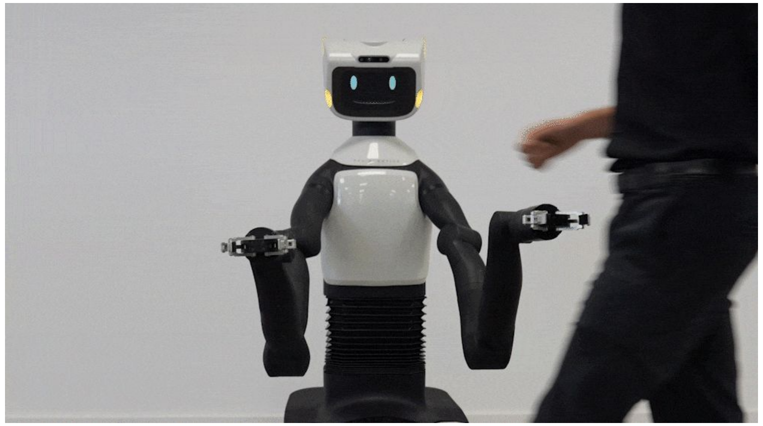
Explore how embodied AI systems learn, act, and collaborate in human environments

Arms to boost **collaboration**

- Designed **arm placement** for seamless human-robot cooperation.
- A **co-design study** to ensure optimal ergonomics and efficiency.
- **Arm mounting flange** with ISO 9409-1 standards, to offer versatility and compatibility.

GRAVITY COMPENSATION

Easily move the robot's arm with precision as it adapts motor force to maintain position



IMPEDANCE CONTROL

Arms are reactive to external disturbances by maintaining the target position as an objective

The **platform enables** natural, safe, controlled physical interaction **across a range of environments**

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Advanced HRI (Human-Robot Interaction) by Design

Conceived as a service robot - meant to work alongside people

Head to boost **expressiveness**

- **Screen** for expressive human-robot interactions.
- Integrated **LEDs** to enhance the visual experience.
- Array of **microphones** and **speakers** to enable smooth dialogue.

Arms to boost **collaboration**

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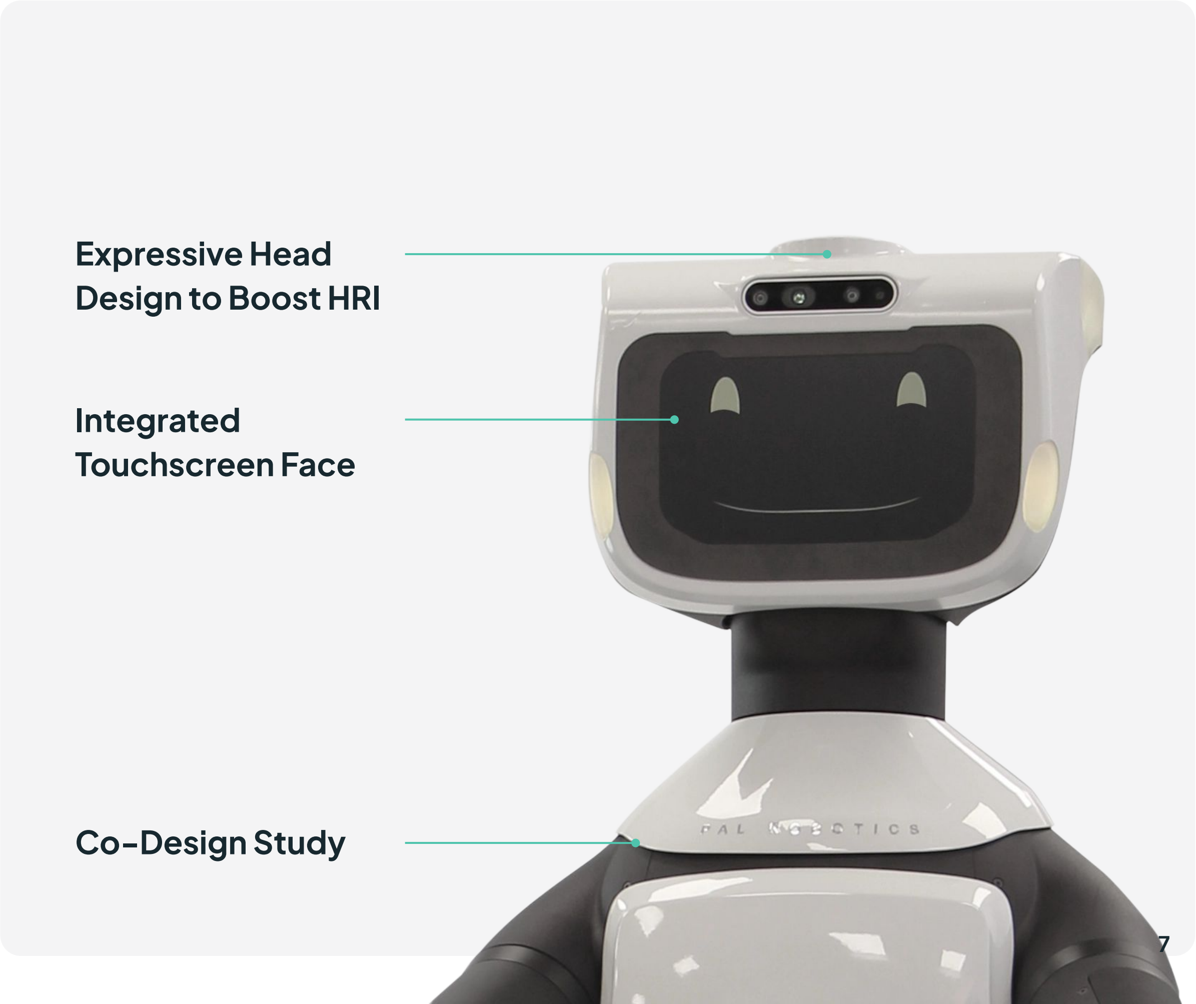

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Flexible Software Architecture

Open-source platform for custom developments

Built on ROS 2 and fully open source, TIAGo Pro offers a flexible software architecture that supports custom development, integration, and experimentation

We designed the Web User Interface to provide you with all the necessary control and management options, to make your experience with TIAGo as smooth as possible! Researchers can easily adapt the platform to test algorithms, interfaces, or behaviors across industries

Multiplatform
access from any device with a Chrome browser

Easy access
find everything on the same browser

Simple to use
enjoy the intuitive interface

Modular
pay only for the features you need

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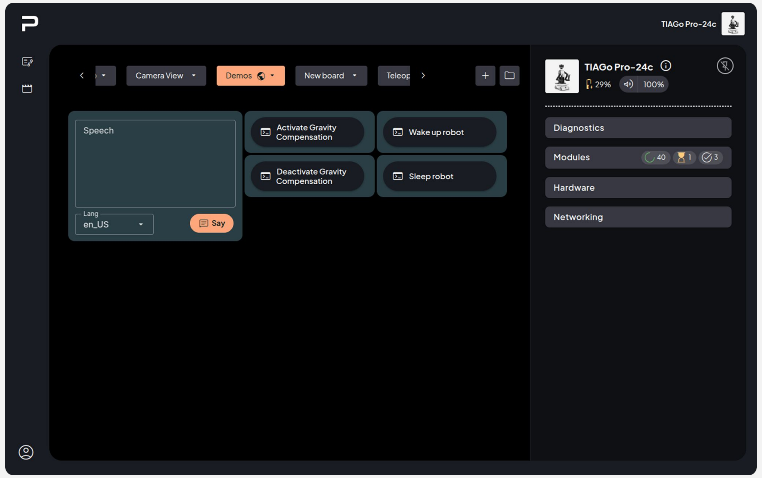
TIAGo Pro WebGUI

A **suite of graphical elements** that facilitate accessing and combining robot capabilities

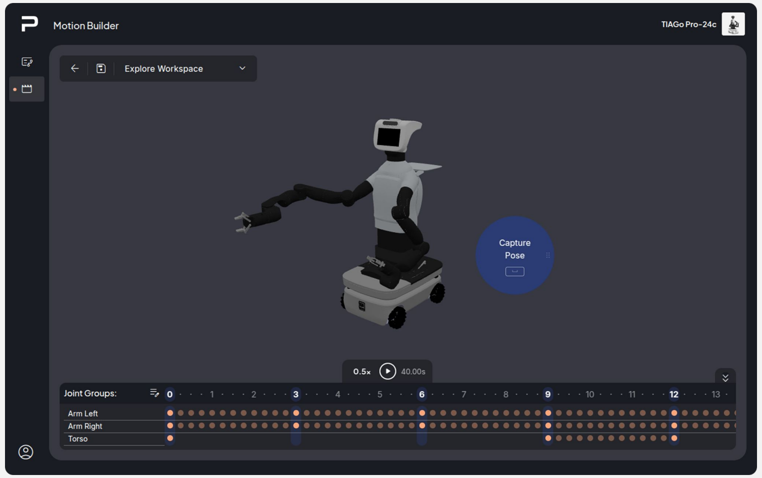
User Boards and Control Center

Real-time robot configuration info and interactive custom boards



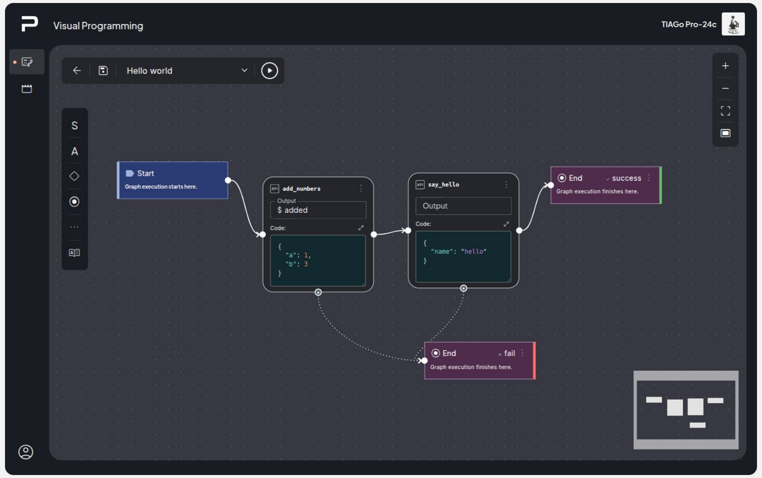
Motion Builder

Motion creation with 3D robot simulation preview



Visual Programming

Drag-and-drop based visual graph for robot programming



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Software Developments

We keep pushing the limits of safe manipulation and HRI

Advanced Navigation Stack

Improved obstacle avoidance
Detection of obstacles using the head RGB-D Camera.

Avoidance of virtual obstacles
User-defined prohibited regions in the map.

Points and regions of interest
Enhanced navigation features.

Impedance Control Stack

Compliant control
Adaptability to external objects forces.

Effort-based
Arm is controlled in effort, closing the loop in position.

Compatible with all default robot functionalities
e.g Joint trajectory Controller and MoveIt.

HRI Human Robot Interaction

Face and emotion recognition
Detects emotions for a database of faces.

Automatic speech recognition and Text to speech
ASR for over 80 languages.

ROS4HRI
The new ROS standard for human-robot interaction.

WBC (Whole Body Control)

Joint Limit Avoidance
The joint values do not go beyond the mechanical limit.

Self-Collision Avoidance
Links in the chain of the controlled joints collide with each other.

Default Pose task
A default reference of arm, head and torso is prescribed.

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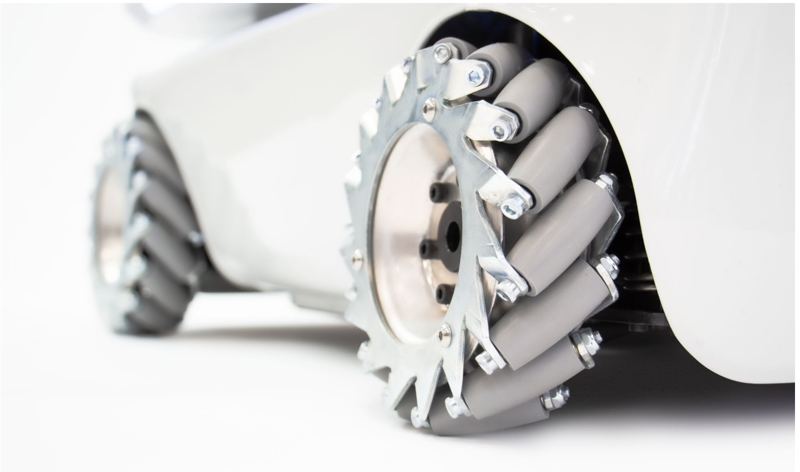

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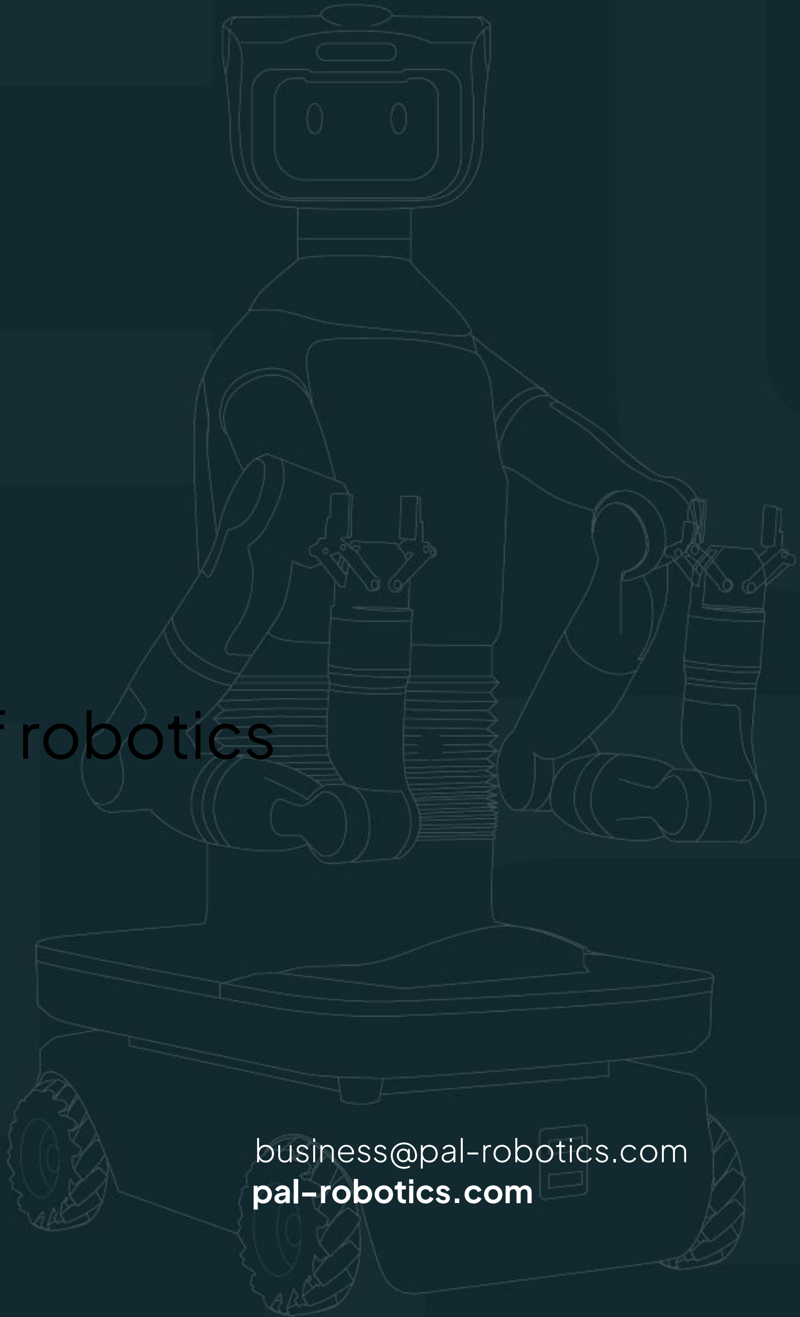
PAL | 20 YEARS OF ROBOTICS

thank you

Let’s continue co-creating the future of robotics



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DESIGNED TO SOLVE YOUR NEEDS

passion for robotics